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Identification of genes for variable regions of immunoglobulins that recognize sialylated glycans

Tetsuya Okuda [a, *], Masahiro Kitamara [b], Shinjiro Kasahara [c], Katsuya Kato [b]

[a] Bioproduction Research Institute, National Institute of Advanced Industrial Science and Technology (AIST), 305-8566 Central 6, 1-1-1 Higashi, Tsukuba, Japan
[b] NGK Spark Plug-AIST Healthcare Materials Cooperative Research Laboratory, National Institute of Advanced Industrial Science and Technology (AIST), 2266-98 Anagahora, Shimoshidami, Moriyama-ku, Nagoya, 463-8560, Japan
[c] NGK Spark Plug Co., Ltd., 2808 Iwasaki, Komaki, Aichi 485-8510, Japan

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ABSTRACT

We previously reported an antibody (clone ID: FR9, IgM-κ) that recognizes the sialyl oligosaccharide Neu5Acα2,6Galβ1,4GlcNAc as an epitope on glycoproteins and glycolipids. In the present study, we developed an antibody (clone ID: AFR45, IgM-κ) that recognizes Neu5Acα2,3Galβ1,4GlcNAc/Glc as an epitope on glycoproteins and glycolipids and compared the nucleotide and amino acid sequences of the immunoglobulin gene variable regions with those of FR9. The heavy chain variable (V_H) regions of FR9 and AFR45 were encoded by different V_H gene segments, each of which was composed of a characteristic D gene segment. The major differences between V_H genes encoding various antibodies deposited in public databases and FR9 and AFR45 were identified in the D gene segment, indicating that D genes play a critical role in determining the epitope specificity of these antibodies. Surprisingly, although FR9 and AFR45 were obtained independently from different mice immunized with different immunogens, the light chain variable (V_L) region nucleotide sequences were identical. The V_L gene consisted of Igkv4-57 and Igkj4 gene segments (Igkv4-57j4), the sequences of which were identical to V_L genes for a number of antibodies against meningococcal group C capsular polysaccharide deposited in public databases. As this polysaccharide is a sialic acid homopolymer, these results indicate that Igkv4-57j4 encodes a V_L common to immunoglobulins that recognize sialylated glycans.

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1. Introduction

Mammalian glycoproteins and glycosphingolipids (GSLs) have a wide variety of glycan structures [1]. As certain glycan structures are associated with cell type, differentiation stage, and pathologic condition, these structures can be used as cell identification markers [1–3]. Sialylation occurs on the nonreducing terminus of mature forms of glycoconjugate glycans [4]. Sialic acid residues are attached to precursor glycans via α2,3, α2,6, and α2,8 linkages, each of which forms a characteristic glycan structure. Differences in the mode of sialic acid linkage affect the properties and functions of the resulting glycoconjugates [4,5]. Thus, technologies that enable researchers to easily distinguish the fine structures of sialylated glycans could contribute to the development of functional glycoproteins and tests to diagnose diseases using glycoconjugate markers. For example, immunoassays, common platforms widely used in medical and pharmaceutical applications [6], could become valuable general-purpose tools for identifying sialylated glycans. However, no practical technology to generate anti-glycan antibodies necessary for the development of such immunoassays is currently available [2,3,7], necessitating further research in this field.

We recently established a method to efficiently induce the production of antibodies against glycans of glycoproteins by immunizing mice with synthetic artificial GSLs (artGSLs) [3,8,9]. Using this technology, we obtained a monoclonal antibody (mAb) (clone ID: FR9, IgM-κ) that recognizes the sialyl oligosaccharide Neu5Acα2,6Galβ1,4GlcNAc (6SLN) [8]. FR9 recognizes the fine structure of 6SLN-containing glycans of glycoproteins and glycolipids, but it does not recognize Neu5Acα2,3Galβ1,4GlcNAc (3SLN)-containing glycans. Thus, characterization of the variable regions of FR9 could provide valuable information regarding how the

* Corresponding author.
E-mail address: t-okuda@aist.go.jp (T. Okuda).

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antibody recognizes the fine structure of sialylated glycans and contribute to the development of practical anti-glycan antibodies.

In this study, we developed a mAb (clone ID: AFR45, IgM-κ) that recognizes Neu5Acα2,3Galβ1,4GlcNAc/Glc as an epitope on glycoproteins and glycolipids and compared the nucleotide and amino acid sequences of the mAb variable regions with those of FR9. The nucleotide sequences encoding the heavy chain variable (V_H) and light chain variable (V_L) regions of these antibodies were compared with those of other antibodies deposited in public databases. Results of homology searches revealed critical differences in the D gene segments of the V_H genes of FR9 and AFR45 in comparison with other antibodies deposited in public databases. In contrast, the nucleotide sequences of genes encoding the V_L regions were conserved for FR9, AFR45, and various antibodies against meningococcal group C capsular polysaccharide (MCPS), which is a sialic acid homopolymer [10,11]. These results indicate that the D genes encoding the V_H regions of these antibodies play a critical role in determining epitope specificity. The V_L gene identified in the present study encodes a V_L common to immunoglobulins that recognize sialylated glycans.

2. Materials and methods

2.1. Materials

ArtGSLs and GM3 (d18:1, C18:0) were synthesized via organic chemistry by Tokyo Chemical Industry Co., Ltd. (Tokyo, Japan) as described previously [9]. The artGSL synthesis procedure is detailed in Japan patent 6143240 and United States patent US10307471B2. Other GSLs, *Salmonella enterica* lipid A, and fetal calf serum fetuin were purchased from Merck (Darmstadt, Germany).

2.2. Immunization and preparation of serum

C3H/HeN mice (CLEA Japan, Tokyo) were used in this study. Mice were immunized with 3SLN-C12S according to a liposome immunization method [9,12]. In brief, 100 μg of artGSL was mixed with 10 μg of lipid A, 0.5 μmol of cholesterol, and 0.5 μmol of dipalmitoylphosphatidylcholine. The mixture was then dissolved in phosphate-buffered saline (PBS) and used as an immunogen. Mice were initially immunized subcutaneously and then intraperitoneally 2 weeks after the first immunization. Serum was prepared from blood collected from the tail vein 7 days after the second immunization. The Committee for Experiments Involving Animals of the National Institute of Advanced Industrial Science and Technology approved all animal experiments, and all experiments were performed in accordance with relevant guidelines and regulations.

2.3. Hybridoma generation

Mice immunized as described above were immunized 3 additional times at 2-week intervals. Three days after the last immunization, splenocytes were collected and fused with mouse Sp2/0-Ag14 myeloma cells (Riken Cell Bank, Tsukuba, Japan) as described previously [13]. Hybridomas were selected using hypoxanthine-aminopterin-thymidine selection medium (RPMI-1640 containing 10% fetal calf serum, 0.1 mM sodium hypoxanthine, 0.4 μM aminopterin, 16 μM thymidine, 10 μg/ml gentamicin, and 5% Briclone [DS Pharma Biomedical, Osaka, Japan]). Cells were cultured at 37 °C in a humidified atmosphere containing 5% CO_2. Culture supernatants were evaluated by enzyme-linked immunosorbent assay (ELISA), and positive clones were selected based on reactivity against 3SLN-C12S and fetuin as an index.

2.4. Characterization of immunoglobulin isotypes

Hybridoma supernatant was diluted 50-fold with 1% bovine serum albumin in PBS, and the isotypes of antibodies present in the supernatant were determined using a mouse monoclonal antibody isotyping kit (Roche Diagnostics GmbH, Mannheim, Germany) according to the manufacturer's instructions.

2.5. ELISA

ELISAs were performed as described previously [9]. In brief, 500 ng of GSLs or 1 μg of glycoprotein was applied to the wells of a 96-well microtiter plate and incubated overnight. After washing twice with PBS, blocking buffer (1% bovine serum albumin in PBS) was added to each well and incubated for 15 min at room temperature, followed by the addition of diluted serum (1:100) or purified mAb (FR9, 1 μg/ml; AFR45, 5 μg/ml). After incubation for 3 h at room temperature, the wells were washed with 0.05% Tween 20 in PBS, followed by addition of a horseradish peroxidase (HRP)-conjugated secondary antibody (anti-IgM). Antibody binding was detected by addition of an HRP substrate (1-Step Ultra TMB-ELISA Substrate; Thermo Fisher Scientific, Waltham, MA, USA) and measurement of the absorbance at 450 nm. For biochemical characterization of the FR9 and AFR45 mAbs, each mAb secreted from the hybridoma into the culture supernatant was purified according to a previously described method using porous zirconia particles [7].

2.6. Determination of the nucleotide sequences of immunoglobulin variable region genes

The nucleotide sequences of immunoglobulin variable region genes were determined by Bio-Peak (Gunma, Japan). Briefly, cDNA was synthesized using total RNA isolated from hybridoma cells, and the V_H and V_L genes were amplified by polymerase chain reaction using degenerate primers and the cDNA as a template. The nucleotide sequences of the V_H and V_L variable region genes were determined by sequencing using the amplified DNA, and the complementarity determining regions (CDRs) in the V_H and V_L genes were identified using the Kabat numbering scheme [14]. Homology searching of target gene sequences against mouse genomic DNA or other gene sequences was performed using BLAST (https://blast.ncbi.nlm.nih.gov/Blast.cgi). Homology analysis of nucleotide and amino acid sequences of target genes was performed using GENETYX-MAC (GENETYX, Tokyo, Japan). The nucleotide sequences reported in this paper were submitted to GenBank under accession numbers LC635721 (V_H of AFR45), LC635722 (V_L of AFR45), LC635723 (V_H of FR9), and LC635724 (V_L of FR9).

3. Results and discussion

3.1. Development of an anti-3SLN mAb and analysis of its biochemical properties

An artGSL containing 3SLN (3SLN-C12S) (Fig. 1A, 3SLN-C12S) was used as an immunogen to generate anti-3SLN antibodies in immunized mice. The chemical structure of 3SLN-C12S is similar to that of GM3 ganglioside (Neu5Acα2,3Galβ1,4Glcβ1,1Cer), a major GSL produced in mammalian cells (Fig. 1A, GM3). The reactivity of serum antibodies isolated from mice immunized with 3SLN-C12S indicated that 3SLN-C12S induced antibodies against this immunogen (Fig. 1B, 3SLN). Antibodies to fetuin, a glycoprotein containing the oligosaccharides 3SLN and 6SLN in its *N*-glycans, were also induced in mice immunized with 3SLN-C12S (Fig. 1B, fetuin). These results indicate that immunizing mice with 3SLN-C12S induces anti-3SLN antibodies.

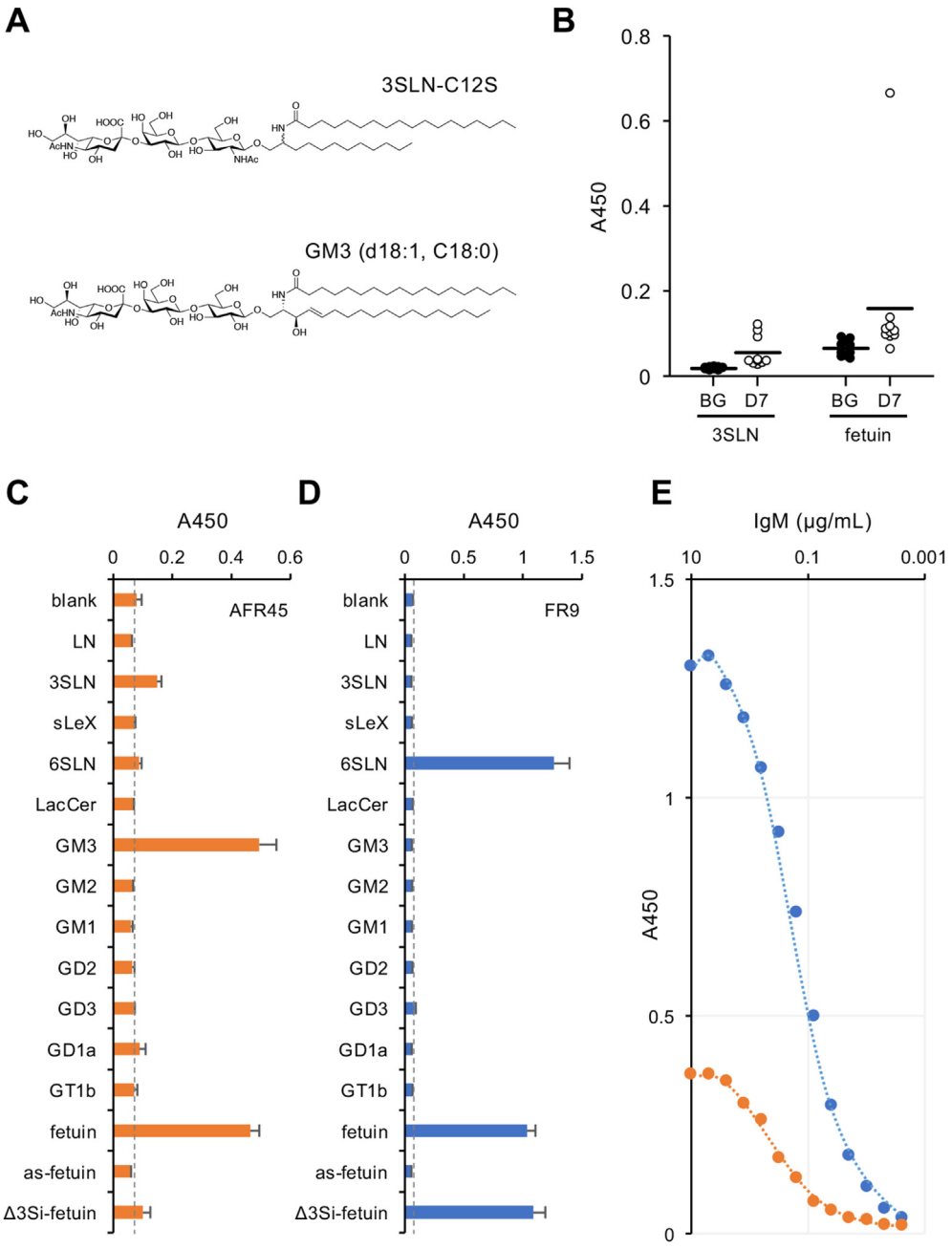


Fig. 1. Characterization of mAb AFR45.
(A) Chemical structure of 3SLN-C12S used for immunization experiment. The ceramide mimetic C12S is composed of a saturated C12-sphingosine mimetic and stearic acid (C18:0). The structure of GM3 (d18:1, C18:0) is shown for comparison with 3SLN-C12S. (B) Antibody-inducing activity of 3SLN-C12S. Mice were immunized with 3SLN-C12S, and serum samples were prepared 7 days after immunization (D7, open circles). The reactivity of serum IgM against 3SLN-C12S (3SLN) or fetuin was analyzed by ELISA (A450), and the reactivity of each individual sample was plotted. BG (closed circles), serum from untreated mice. Average absorbance values are indicated as horizontal bars. (C) Reactivity of mAb AFR45 against several glycoconjugates as evaluated by ELISA. LN, 3SLN, sLeX, and 6SLN indicate the conjugates of these oligosaccharides and previously reported ceramide mimetics [9]. (D) Reactivity of mAb FR9 against several glycoconjugates as evaluated by ELISA. (E) Reactivity of various concentrations of AFR45 (orange circle) and FR9 (blue circle) against fetuin as evaluated by ELISA. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Splenocytes isolated from a 3SLN-C12S–immunized mouse in which the serum contained antibodies that strongly reacted with fetuin were used for hybridoma preparation. The prepared hybridomas were screened for the production of antibodies that reacted with fetuin and 3SLN-C12S as indexes, resulting in the isolated of a hybridoma (clone ID: AFR45) that produced anti-3SLN IgM-κ. Analysis of the reactivity and binding specificity of mAb AFR45 against several glycoconjugates by ELISA showed that AFR45 reacted with 3SLN-C12S, GM3, and fetuin but not other GSLs, asialofetuin, or fetuin treated with α2,3-sialidase (Fig. 1C). AFR45 preferentially reacted with GM3 and fetuin containing Neu5Acα2,3Galβ1,4GlcNA/Glc in the non-reducing end and did not react with GD1a or GT1b, which contain Neu5Acα2,3Galβ1,3GalNAc in the nonreducing end. A unique property of AFR45 was that it reacted more strongly with GM3 and fetuin than the immunogen, 3SLN-C12S. These results indicate that the molecular structure of the reducing end of the epitope oligosaccharide affects the reactivity of AFR45 with the epitope.

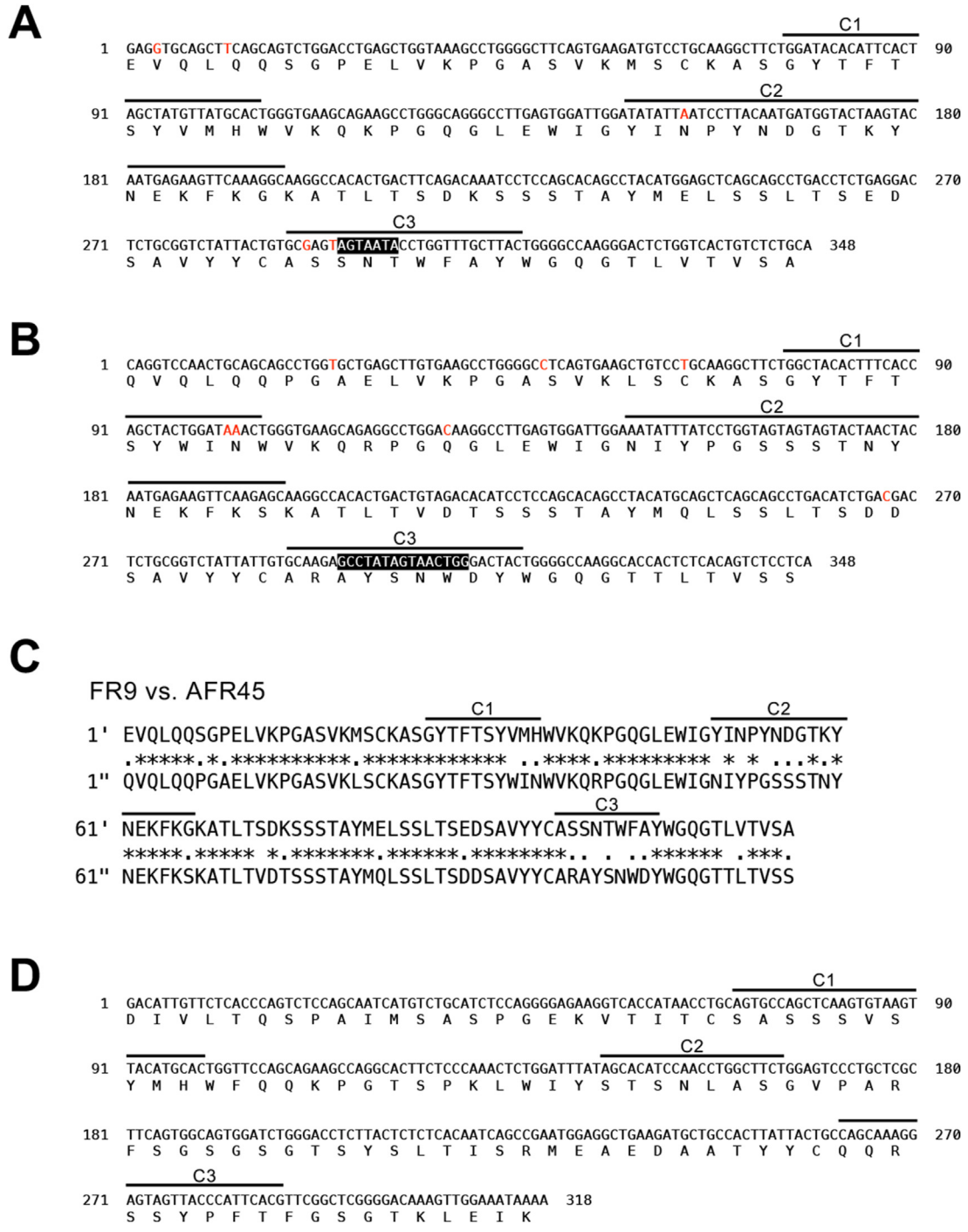


Fig. 2. Nucleotide and amino acid sequences of the variable region of FR9 and AFR45.
V_H gene nucleotide sequence and inferred amino acid sequence of FR9 (A) and AFR45 (B). Bases that are mutated compared to mouse germline V_H sequences are shown in red font. The D gene segment of each V_H sequence is highlighted by a closed box. C1, C2, and C3 with underline indicate CDR1, CDR2, and CDR3, respectively. (C) Homology analysis of V_H amino acid sequences of FR9 and AFR45. Asterisks and dots represent conserved amino acids and amino acids with the same property, respectively. (D) Nucleotide sequence and inferred amino acid sequence of Igkv4-57j4. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

The mAb FR9, which specifically reacted with 6SLN in glycoconjugates (Fig. 1D), was isolated from a mouse immunized with a 6SLN-containing artGSL as described previously [8]. This antibody reacted strongly with fetuin even at low concentrations (Fig. 1E) and exhibited greater specificity for the epitope (Fig. 1D). We also reported previously that FR9 did not react with oligosaccharides with structures very similar to 6SLN, such as Neu5Acα2,6Galβ1,4Glc [7,8].

3.2. Nucleotide and amino acid sequences of FR9 and AFR45 variable regions

cDNAs prepared from total RNA isolated from FR9 and AFR45 hybridomas were used to determine the nucleotide and amino acid sequences of the immunoglobulin variable regions. Results of BLAST searches of public databases indicated that the V_H of FR9 was encoded by Ighv1-14 and Ighj3 V_H gene segments and a D gene

Table 1
Homology between amino acid sequences of the CDRs of anti-sialyl glycan and anti-MCPS mAbs.

Clone	CDR1	CDR2	CDR3	Full length
(**A**) vs. FR9				
AFR45	70.0%	52.9%	22.2%	74.1%
177.16	80.0%	70.6%	30.0%	78.6%
4-2-C	80.0%	70.6%	30.0%	78.6%
2016.3	70.0%	70.6%	30.0%	74.4%
(**B**) vs. AFR45				
177.16	60.0%	47.1%	40.0%	70.3%
4-2-C	60.0%	47.1%	30.0%	69.5%
2016.3	50.0%	52.9%	20.0%	65.8%

Table 1 shows the percent homology between the amino acid sequences of the V_H CDRs of various antibodies and FR9 (**A**) or AFR45 (**B**). Amino acid sequences are shown in Fig. 3B and Supplementary Fig. S1.

segment characteristic of FR9 (Fig. 2A, highlighted with closed box). A cDNA sequence excluding the D region exhibited 98.5% identity with the mouse germline gene. The mutated bases are shown in red font in Fig. 2A, and some of the mutated bases were found in the CDR. Major differences between this sequence and the V_H gene sequences of various antibodies deposited in public databases were observed in the D gene segment. The V_H of AFR45 is encoded by Ighv1-62 and Ighj2 V_H gene segments and a D gene segment characteristic of AFR45 (Fig. 2B, highlighted with closed box). A cDNA sequence excluding the D gene segment exhibited 97.9% identity with the mouse germline gene. The mutated bases are shown in red font in Fig. 2B, and some of the mutated bases were found in the CDR. Major differences between this sequence and V_H gene sequences of various antibodies deposited in public databases were also observed in the D gene segment. The cDNA and amino acid sequences of the V_H regions of FR9 and AFR45 exhibited 81.3%

A

Igkv4-57j4 vs. 177.16

```
                                C1                                C2
 1' DIVLTQSPAIMSASPGEKVTITCSASSSVSYMHWFQQKPGTSPKLWIYSTSNLASGVPAR
    .************************************************************
 1" QIVLTQSPAIMSASPGEKVTITCSASSSVSYMHWFQQKPGTSPKLWIYSTSNLASGVPAR
                                C3
61' FSGSGSGTSYSLTISRMEAEDAATYYCQQRSSYPFTFGSGTKLEIK
    **************************************************
61" FSGSGSGTSYSLTISRMEAEDAATYYCQQRSSYPFTFGSGTKLEIK
```

Igkv4-57j4 vs. 4-2-C

```
                                C1                                C2
 1' DIVLTQSPAIMSASPGEKVTITCSASSSVSYMHWFQQKPGTSPKLWIYSTSNLASGVPAR
    **..********************************************************
 1" DILMTQSPAIMSASPGEKVTITCSASSSVSYMHWFQQKPGTSPKLWIYSTSNLASGVPAR
                                C3
61' FSGSGSGTSYSLTISRMEAEDAATYYCQQRSSYPFTFGSGTKLEIK
    **************************************************
61" FSGSGSGTSYSLTISRMEAEDAATYYCQQRSSYPFTFGSGTKLEIK
```

Igkv4-57j4 vs. 2016.3

```
                                C1                                C2
 1' DIVLTQSPAIMSASPGEKVTITCSASSSVSYMHWFQQKPGTSPKLWIYSTSNLASGVPAR
    .************************************************************.*
 1" QIVLTQSPAIMSASPGEKVTITCSASSSVSYMHWFQQKPGTSPKLWIYSTSNLASGVPTR
                                C3
61' FSGSGSGTSYSLTISRMEAEDAATYYCQQRSSYPFTFGSGTKLEIK
    **********.**********************.***.*******
61" FSGSGSGTSYTLTISRMEAEDAATYYCQQRSSYPYTFGGGTKLEIK
```

B

```
Clone   CDR1                            CDR2                                    CDR3
FR9     GGATACACATTCACTAGCTATGTTATGCAC  TATATTAATCCTTACAATGATGGTACTAAGTACAATGAGAAGTTCAAAGGC  GCGAGTAGTAATACCTGG---TTTGCTTAC
        G  Y  T  F  T  S  Y  V  M  H    Y  I  N  P  Y  N  D  G  T  K  Y  N  E  K  F  K  G   A  S  S  N  T  W  -  F  A  Y
AFR45   GGCTACACTTTCACCAGCTACTGGATAAAC  AATATTTATCCTGGTAGTAGTAGTACTAACTACAATGAGAAGTTCAAGAGC  GCAAGAGCCTATAGTAAC---TGGGACTAC
        G  Y  T  F  T  S  Y  W  I  N    N  I  Y  P  G  S  S  S  T  N  Y  N  E  K  F  K  S   A  R  A  Y  S  N  -  W  D  Y
177.16  GGATACACATTCACTGAATACACCATGCAC  GGTATTAATCCTAACAATGGTGGTACTAGGTACAACCAGAAGTTCAAGGGC  GCAACACGTTCCAACTGGTACTTCGATGTC
        G  Y  T  F  T  E  Y  T  M  H    G  I  N  P  N  N  G  G  T  R  Y  N  Q  K  F  K  G   A  T  R  S  N  W  Y  F  D  V
4-2-C   GGATACACATTCACTGAATACACCATGCAC  GGTATTAATCCTAACAATGGTGGTACTAGATACAACCAGAAGTTCAAGGGC  GCAGATAGGTACGTCTGGTACTTCGATGTC
        G  Y  T  F  T  E  Y  T  M  H    G  I  N  P  N  N  G  G  T  R  Y  N  Q  K  F  K  G   A  D  R  Y  V  W  Y  F  D  V
2016.3  GGAAACACATTCACTGAATACACCATGCAC  GGTATTAATCCTAAAAATGGTGGTACTAACTACAACCAGAAGTTCAAGGGC  GTAAGCCGTAGCGTCTGGTACTTCGATGTC
        G  N  T  F  T  E  Y  T  M  H    G  I  N  P  K  N  G  G  T  N  Y  N  Q  K  F  K  G   V  S  R  S  V  W  Y  F  D  V
```

Fig. 3. Amino acid sequence homology between the variable regions of FR9, AFR45, and anti-MCPS antibodies.
(A) Analysis of V_L amino acid sequence homology between Igkv4-57j4 and anti-MCPS antibodies. C1, C2, and C3 with underline indicate CDR1, CDR2, and CDR3, respectively. Asterisks and dots represent conserved amino acids and amino acids with the same property, respectively. GenBank accession numbers for the V_L sequence of 177.16 (IgG1-κ), 4-2-C (IgG3-κ), and 2016.3 (IgG3-κ) are AY229942, AY639148, and AY229956, respectively. (B) Amino acid sequences of V_H CDRs of FR9, AFR45, and anti-MCPS antibodies (177.16, 4-2-C, 2016.3). Table 1 summarizes the percent homology of the CDR sequences. GenBank accession numbers for the V_H sequence of 177.16, 4-2-C, and 2016.3 are AY229941, AY639149, and AY229955, respectively.

and 74.1% homology, respectively (Fig. 2C and Table 1). Surprisingly, although the FR9 and AFR45 antibodies were obtained independently from different mice immunized with different immunogens, the sequences of the V_L genes were identical (Fig. 2D). The V_L gene was encoded by Igkv4-57 with Igkj4 V_L gene segments (Igkv4-57j4), and the nucleotide sequence was identical to that of the mouse germline gene.

3.3. Comparative analysis of variable regions of immunoglobulins recognizing sialylated glycans and MCPS

The results of further homology searches against various V_L genes deposited in public databases revealed that the gene Igkv4-57j4 encodes the V_L region for a number of antibodies against MCPS, a linear homopolymer of α2,9-linked sialic acid [10,11]. Three representative anti-MCPS mAbs clones [11], 177.16 (IgG1-κ), 4-2-C (IgG3-κ), and 2016.3 (IgG3-κ), are shown in Fig. 3. Although the amino acid sequences encoded by these V_L genes were found to be highly conserved, exhibiting 95–100% homology with Igkv4-57j4, these V_H exhibited a low degree of similarity (approximately 70%) with the corresponding sequences for FR9 and AFR45 (Table 1, full length).

A further homology analysis of these mAbs revealed that the CDRs of the AFR45 V_H exhibited lower similarity with the CDRs of the anti-MCPS mAbs than did those of FR9 (Fig. 3B and Table 1). CDR1 and CDR2 in the V_H regions of the anti-MCPS mAbs exhibited 70–80% and 70% similarity, respectively, with those in FR9 and 50–60% and 40–50% similarity, respectively, with those in AFR45. Furthermore, CDR2 in the AFR45 V_H exhibited low similarity with CDR2 in the FR9 V_H (52.9%, Table 1). As CDR3 contains a D gene segment unique to each mAb, the homologies of CDR3 were very low across all mAbs (approximately 30%). These results indicate that the structure of the epitope recognition domain of AFR45 differs significantly from that of the other antibodies, suggesting that antibodies recognize α2,3-sialic acid in a manner distinct from that of antibodies that recognize α2,6- and α2,9-linked sialic acids.

A BLAST homology search using Igkv4-57j4 did not return any hits for V_L genes encoding mAbs against meningococcal group B capsular polysaccharide, a linear homopolymer of α2,8-linked sialic acid [11]. These results indicate that the Igkv4-57j4 commonly encodes the V_L regions of immunoglobulins that react with glycans containing α2,3-, α2,6-, or α2,9-lined sialic acid as an epitope. Our data also suggest that diversity among the V_H genes of these mAbs facilitates recognition of the fine structure of these epitope glycans.

In this study, we identified Igkv4-57j4 as a gene that commonly encodes the V_L regions of immunoglobulins that recognize sialylated glycans as epitopes. The biochemical properties of antibodies in which the V_L regions are encoded by Igkv4-57j4 indicate that this V_L plays a role in antibody recognition of α2,3-, α2,6-, and α2,9-linked sialic acids. However, FR9 and AFR45 did not react with GD3, a GSL containing α2,8-linked sialic acid (Fig. 1). In addition, Igkv4-57j4 was not identified as the V_L gene for immunoglobulins against meningococcal group B capsular polysaccharide, a linear homopolymer of α2,8-linked sialic acid [11]. Thus, the V_L region encoded by Igkv4-57j4 does not play a role in recognizing α2,8-linked sialic acid. Based on these observations, we conclude that the Igkv4-57j4 V_L gene could be used in the development of antibodies that specifically react with sialylated glycans containing α2,3-, α2,6-, or α2,9-linked sialic acid.

The nucleotide sequences of the V_L genes for FR9/AFR45 were identical to the counterpart sequences in mouse genomic DNA, and only a few mutated bases were found in the V_L gene nucleotide sequences for the anti-MCPS antibodies. Due to this low rate of mutations, almost no amino acid sequence alterations were observed among the V_L regions of the anti-MCPS antibodies. This result indicates that mammals harbor V_L genes that encode immunoglobulins that recognize sialylated glycans. Although healthy humans [15,16] and mice [13] are known to constitutively produce anti-mammalian glycan antibodies, why these antibodies (which are also autoantibodies) are induced remains unclear. An important aspect of our findings is that mammals harbor V_L genes that encode immunoglobulins that recognize capsular polysaccharides of pathogenic bacteria, and these V_L genes also encode immunoglobulins that recognize mammalian glycans. This finding strongly suggests that endogenous anti-mammalian glycan antibodies are induced in mammals in association with host defense against pathogenic bacteria.

Our findings also revealed that a subset of B cells that recognize sialylated glycans as epitopes can be characterized by expression of the Igkv4-57j4 gene. This genetic information can contribute to the analysis of immunoglobulins with as yet undetermined epitopes. Furthermore, this genetic information could serve as a useful index for the diagnosis of meningococcal infections and autoimmune diseases involving anti–sialylated glycan antibodies, such as Guillain-Barré syndrome [17].

Declaration of competing interest

The National Institute of Advanced Industrial Science and Technology holds patents (JP6172687, US9796785B2, and EP2993184B1) that relate to the materials reported in this article. The authors declare no other competing interests.

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Abbreviations

V_H	heavy chain variable (region)
V_L	light chain variable (region)
GSL	glycosphingolipid
artGSL	artificial glycosphingolipid
mAb	monoclonal antibody
3SLN	Neu5Acα2,3Galβ1,4GlcNAc
6SLN	Neu5Acα2,6Galβ1,4GlcNAc
MCPS	meningococcal group C capsular polysaccharide
CDR	complementarity determining region
LN	Galβ1,4GlcNAc
sLeX	Neu5Acα2,3Galβ1,4(Fucα1,3)GlcNAc
LacCer	Galβ1,4Glcβ1,1Ceramide
GM3	Neu5Acα2,3Galβ1,4Glcβ1,1Ceramide
GM2	GalNAcβ1,4(Neu5Acα2,3)Galβ1,4Glcβ1,1Ceramide
GM1	Galβ1,3GalNAcβ1,4(Neu5Acα2,3)Galβ1,4Glcβ1,1Ceramide
GD2	GalNAcβ1,4(Neu5Acα2,8Neu5Acα2,3)Galβ1,4Glcβ1,1Ceramide
GD3	Neu5Acα2,8Neu5Acα2,3Galβ1,4Glcβ1,1Ceramide
GD1a	Neu5Acα2,3Galβ1,3GalNAcβ1,4(Neu5Acα2,3)Galβ1,4Glcβ1,1Ceramide
GT1b	Neu5Acα2,3Galβ1,3GalNAcβ1,4(Neu5Acα2,8Neu5Acα2,3)Galβ1,4Glcβ1,1Ceramide
as-fetuin	asialo-fetuin
Δ3Si-fetuin	α2,3-linked sialic acids-removed fetuin

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbrc.2021.06.039.

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